

E. Coli Heat Shock Transformation

Heat shock transformation is a common method used to introduce plasmid DNA into chemically competent E. coli cells. Competent cells are prepared to have permeable membranes that can take up DNA under specific conditions. During transformation, the plasmid DNA is mixed with competent cells and briefly exposed to a sudden increase in temperature (typically 42 °C). This heat shock creates a thermal imbalance across the cell membrane, driving the uptake of DNA into the cytoplasm. The cells are then allowed to recover in nutrient-rich media to express the antibiotic resistance gene carried by the plasmid before being plated on selective agar.

Protocol adapted from [High Efficiency Transformation Protocol \(NEB #C2987H/#C2987I\)](#)

Reagents

✓	Name	Quantity	Location/Notes
☐	Prepared chemically-competent E. coli	200 µL aliquot	Stored at -80C
☐	Plasmid DNA	~100ng, 1-5uL	
☐	LB media		
☐	LB agar plate with antibiotic		

Equipment

✓	Name	Quantity	Location/Notes
☐	Centrifuge		
☐	Water bath (42°C)		
☐	1 mL tube	1	

☐ Incubator (37°C)☐ Ice☐ Spreaders

Instructions

1. Thaw competent cells on ice for 10 mins.
2. Transfer to centrifuge tubes
3. Add 1-5 μ L containing 1 pg-100 ng of plasmid DNA to cells and mix.
Do not vortex.
 - a. Consider including a negative control with digested vector only to assess effect of any background intact empty plasmid in plasmid assembly mix
4. Place the tube on ice for 30 mins, **do not mix**.
5. Heat shock at 42°C for 45s exactly by placing into waterbath
6. Place on ice for 5 mins, **do not mix**.
7. Add 950 μ L LB media and place in a shaking incubator (250rpm) for 1 hour.
8. Spin down cell mixture in centrifuge, discard supernatant leaving around 100 μ L, resuspend in remaining liquid.
9. Spread 100 μ L of cell mixture onto the LB selection plates. Incubate at 37°C overnight then observe plate for transformant colonies.

Notes

- NEB says to heat shock for 30s but we aren't using the provided transformation tubes

References

- Last year's protocol
- Protocol adapted from [High Efficiency Transformation Protocol \(NEB #C2987H/#C2987I\)](#)